



Homevoters vs. leasevoters: A spatial analysis of airport effects



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ABSTRACT

We use a public referendum on a new aviation concept in Berlin, Germany, as a natural experiment to analyze how the interaction of tenure and capitalization effects influences the outcome of direct democracy processes. We distinguish between homevoters, i.e., voters who are homeowners, and leasevoters, i.e., voters who lease their homes. We expect that homevoters would be more likely to support initiatives that positively affect the amenity value of a neighborhood because some of the related benefits of leasevoters are neutralized by adjustments in market rents. Likewise, homevoters would be more likely to oppose initiatives that negatively affect the amenity value of a neighborhood. Our empirical results are consistent with these expectations, implying that public votes on local public goods do not necessarily reflect the spatial distribution of welfare effects in mixed-tenure environments.

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1. Introduction

A central prediction of the standard theoretical urban economics framework is that any advantage that is specific to a particular location must be offset by a correspondingly higher rent or purchasing price of a property (e.g., [Alonso, 1964](#); [Roback, 1982](#); [Rosen, 1974](#)). To homeowners, an increase in the quality of location implies a windfall capital gain. During their remaining time in the home, homeowners receive those benefits as imputed rents as they benefit from an increase in service flow. Upon selling their home, homeowners receive the discounted flow of the remaining future benefits through a higher sale price. Homevoters, i.e., voters who are homeowners, thus unambiguously benefit from local improvements in location quality and are therefore likely to support

initiatives that induce positive amenities. This is the homevoter hypothesis ([Fischel, 2001](#)).

For renters in unregulated markets, however, the capitalization effect negatively compensates for an increase in service flow because of improvements in location quality. If a positive amenity change attracts households with stronger preferences for that amenity, the increase in rent will exceed the benefit sitting renters derive from the increase in service flow. In this case, the net effect on renters who stay put will be negative. Leasevoters, i.e., voters who lease their homes, may therefore be indifferent to or oppose initiatives that induce positive amenity.

If sitting renters are protected from rent increases due to rent control, they will enjoy the increase in service flow at no additional cost during the remaining tenure. However, unlike homeowners, renters do not receive a windfall capital gain when they vacate their home. Thus, even with rent control, homeowners derive higher net benefits from improved local amenities than renters do, though not to the same extent as in a free market. Leasevoters, in any case, are less likely to support initiatives that enhance local amenities than are homevoters.

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Several studies suggest that projected house price capitalization effects significantly influence the degree of support for public initiatives and projects (Ahlfeldt, 2011; Brunner and Sonstelie, 2003; Brunner et al., 2001; Dehring et al., 2008; Hilber and Mayer, 2009). More generally, the literature on the political economy of housing markets suggests a strong link among the nature of the political process, the ownership of land, and patterns of development (Baskaran, 2012; Cellini et al., 2010; DiPasquale and Glaeser, 1999; Kahn, 2011; Millard-Ball, 2012; Solé-Ollé and Viladecans-Marsal, 2013). Our contribution enriches this literature by explicitly differentiating between homevoter and leasevoter behavior in a public poll. We do so by examining an interaction effect between tenure (being a homeowner or renter) and the anticipated capitalization effect.

We apply our test of the homevoter hypothesis to the 2008 public referendum on the Tempelhof Airport in Berlin, Germany. Partly because of its history, Berlin possessed three relatively small airports in the early 1990s. Tegel Airport and Tempelhof Airport are centrally located within the boundaries of former West Berlin, whereas Schoenefeld Airport lies close to the southeastern boundary of Berlin and served East Berlin during the division period. On July 4, 1996, in a so-called *Konsensbeschluss* (“consensus decision” hereafter), the Prime Minister of Brandenburg, the Governing Mayor of Berlin, and the Federal Minister for Transport decided to redevelop Schoenefeld Airport into a large-scale international hub airport, named Berlin–Brandenburg International Airport, where all air traffic would be concentrated (Abgeordnetenhaus von Berlin, 1996). While alternative locations outside Berlin would have offered a more sparsely populated environment, Schoenefeld’s key advantages were the relatively short distance to the center of Berlin and the existing infrastructure (a runway and access to the highway and transit network).

The Tempelhof closure was scheduled for October 31, 2008. As this date approached, the intensity of the protests against the plan steadily increased. Opposition to the closure was stronger for Tempelhof than for Tegel because the former closure was imminent, whereas Tegel was announced to remain in operation until the opening of Berlin–Brandenburg International. In addition, Tempelhof was the object of emotional attachment because of its role as an important access point for the airlift during the 1948–1949 Berlin Blockade. Because the closures of Tempelhof and Tegel were legally binding requirements for the opening of the new major airport at the Schoenefeld site, the Tempelhof referendum was, in fact, a referendum on the new aviation concept as a whole. The referendum was held on April 27, 2008, and was approved by a majority of those who voted, but it failed to achieve the minimum favorable vote quorum of 25 percent YES votes (against the new aviation concept) at the total electorate.¹

The Berlin Tempelhof referendum provides a particularly interesting natural experiment to test the homevoter hypothesis. First, the referendum provides us with rich spatial variation in local costs and benefits. It was directly or indirectly connected to three airports whose location has been an artifact of the particular history of the city. Second, the Berlin housing market exhibits a large degree of spatial variation in terms of its tenure structure. Third, unlike many US markets, the German rental market is strongly regulated. The extent to which landlords can pass on increases in market rents to incumbent residents is limited. Renters are largely hedged against rent increases as long as they do not move out.² Because renters will therefore be less inclined to oppose initiatives

that bring positive local amenities, the robust evidence that we document on homevoter and leasevoter effects in this institutional context provides particularly strong support for the empirical relevance of the homevoter hypothesis.

After introducing our data in the next section, we proceed with our analysis in two major steps. In the first step, we estimate the property price effects associated with the new aviation concept (Section 3). Following Dehring et al. (2008), we argue that past announcement effects provide a noisy signal to homevoters and leasevoters. In the second step, we analyze homevoter and leasevoter effects by linking the voting outcome to the estimated price signal and the homeownership rate (Section 4). Section 5 presents extensions and robustness checks of the baseline models. The final section concludes the study.

2. Data

Our test of the homevoter hypothesis requires a variety of data at a spatially disaggregated level: first, the share of NO votes in the Tempelhof referendum, from which we infer the local support for the new aviation concept; second, measures of exposure to aircraft noise and accessibility to airport terminals to approximate the expected change in local amenity value; third, a set of socioeconomic voter characteristics to control for correlated preferences that impact voting decisions but are unrelated to local changes in the amenity value; fourth, a proxy of the proportion of voters that belong to home-owning households to distinguish between homevoters and leasevoters; and fifth, a comprehensive property transactions data set to estimate the announcement effect of the new aviation concept which serves as a measure of the price effects voters associate with the new concept. We describe the sources and the processing of the data in more detail below.

2.1. Spatial unit of analysis

The voting precincts form our main analysis unit according to which all other data were organized using a geographic information system (GIS) and the framework of the Urban and Environmental Information System of the Department for Urban Development and the Environment of the Berlin State government administration (Senatsverwaltung für Stadtentwicklung Berlin, 2006). In merging the data, we compute the precinct values as weighted averages of the spatial units that overlap with a given precinct, with the weights being proportionate to the respective shares of the geographic area of the precinct (Arntz and Wilke, 2007; Goodchild and Lam, 1980). All distance computations were made using this GIS framework with reference to precinct centroids.

2.2. Voting data

The data on the voting results for the Tempelhof referendum were obtained from the Statistical Office for Berlin–Brandenburg. Of the 881,035 votes that were cast, 650,464 votes were cast in person at the ballot box and can be used in the empirical analyses. These results are available as aggregated outcomes at the level of 1201 voting precincts and are merged with an electronic map of the precinct boundaries via a unique identifier variable.

The remaining votes were cast by mail (postal votes) and cannot be considered because a similarly detailed geo-reference is not available. The highest spatial disaggregation at which the voting outcome can be obtained separately for postal voters comprises the 12 city districts (Bezirke). Across Bezirke, the correlation coefficient between the share of NO votes among individuals who voted in person and the share of NO votes among all voters (which includes voters who voted in person and postal voters) is as high as

¹ More detailed information on the history of Berlin’s airports and the Tempelhof referendum is provided by Nitsch (2009).

² Housing rental law in Germany is highly regulated, complex, and biased toward renters. Landlords can only freely negotiate the rent when a new rental contract is signed. See Westerheide (2011) for details.

0.9998. The correlation coefficient between the share of NO votes among voters who voted in person and the share of NO votes among postal voters is similarly high (0.9973). It is therefore unlikely that the omission of postal voters introduces substantial sample selection bias into our data.

2.3. Noise data

The Department for Urban Development and the Environment provided detailed information on exposure to aircraft noise from Tegel airport in 2007 at the level of 10×10 -m grid cells. Noise levels are recorded for all developed properties within the Tegel air corridor and are expressed using a long-term sound pressure index (L_{den}) that is equivalent to the standard log-decibel scale. For Tempelhof, similar data were obtained from the Berlin airport's operating company.³ For Schoenefeld, which lies outside the boundaries of Berlin and whose air corridor only partly crosses Berlin's territory, the best available information was a map of the noise abatement zone (Laerschutzzone), which anticipates the expected increase in noise levels upon the opening of the new Berlin–Brandenburg International Airport.

Fig. 1 shows the locations of the three airports relative to the central business district (CBD) and illustrates the areas where aircraft noise levels exceed 45 decibels. We define all of the aircraft noise variables such that they express the number of decibels above the 45-decibel threshold; for example, values of 5, 10, and 15 excess decibels correspond to 50, 55, and 60 decibels, respectively. Values below 45 decibels are set to zero excess decibels to reflect levels of aircraft noise that are indistinguishable from other types of noise (Ahlfeldt and Maennig, 2013).

2.4. Socioeconomic characteristics

Detailed information on socio-demographic characteristics, including population, age groups, proportions of males and non-German individuals (for 15,937 blocks), unemployment rate (for 338 traffic cells), and the outcome of the 2006 state elections (for 2424 small voting precincts) were available from the statistical office in Berlin. We complemented these data with purchasing power per capita estimates (available for 197 post codes) derived from a report by the Consumer Research Society (Gesellschaft für Konsumforschung). All data refer to 2008. To account for the possibility that the residents of the former East Berlin feel less attached to a West Berlin airport, we include in the vector of location variables an indicator variable that is set to one for precincts within the boundaries of the former East Berlin.

2.5. Homeownership rate

The last full population census in Germany took place in 1987. Unlike in the U.S., official data on the homeownership rate at a highly disaggregated spatial level are therefore not readily available in Germany. To approximate the homeownership rate at the precinct level, we exploit a particularity of the Berlin housing market: the segmentation of the market into (1) detached, semidetached, and attached single-family houses, villas, and townhouses that are almost entirely occupied by owners and (2) typical downtown apartment buildings, which are usually five stories tall and are almost entirely occupied by renters.

The tendency of multifamily units to be renter occupied and of single-family units to be owner occupied is not exclusive to the

housing market in Berlin.⁴ Using 2011 city-level data provided by the Federal Statistical Office of Germany (available at www.destatis.de), we find a high correlation between the shares of owner-occupied dwellings and the share of one- and two-family houses in the total stock across the 100 largest German cities in Fig. 2. The correlation coefficient is as high as 0.89. In Table 1, we compare the shares of owner-occupiers in one- and two-family houses as well as in buildings with three or more families, in the US, Germany, and Berlin, using data from the US Integrated Public Use Microdata Series (Ruggles et al., 2010, available at www.usa.ipums.org) and the German Microcensus (available at www.forschungsdatenzentrum.de).⁵ The owner-occupancy rates by dwelling type are remarkably similar in the US and Germany. Close to 80 percent of one- and two-family houses are owner occupied, whereas more than 80 percent of the dwellings with three or more families are inhabited by renters. With similar rates exceeding 90 percent, the degree of segregation is even higher in Berlin.⁶

The Department for Urban Development and the Environment categorizes the building structure in Berlin into 15,937 largely homogenous statistical blocks (statistische Bloecke). Using this information, we approximate the homeownership rate (H) for a voting precinct as the proportion of the total population above the age of 18 (the electorate) that lives within the boundaries of a statistical block of one- or two-family houses. To test this procedure, we compute this measure for the 12 city districts (Bezirke). We then compare the results to observed Bezirke-level owner-occupancy rates reported by the Investitionsbank Berlin, the development bank of the Federal State of Berlin (IBB, 2008). As shown in Fig. 3, the two measures are approximately linearly related. The R^2 of the linear fit is 0.75. In most districts, the share of population above the age of 18 living in one- or two-family houses is larger than the reported proportions of owner-occupied dwellings. We consider this feature of our approximation of the homeownership rate to be positive because it accounts for the tendency of households living in one- or two-family houses to be relatively large in size. The proportion of “homevoters” should normally be expected to be larger than that suggested by the share of owner-occupied dwelling stock in the total stock.

As with any imputation of critical variables, our procedure has potential costs. For instance, some measurement error in our approximation of the homeownership rate could lead to attenuation bias in our estimated homevoter and leasevoter effects. Because the segregation of homeowners and renters into different types of dwellings is so striking in Berlin and because we observe areas of homogenous dwelling characteristics and population at a level as detailed as housing blocks, we expect any measurement error effects to be limited. All qualitative conclusions should remain unaffected.

2.6. Property data

We observe all transactions of built-up parcels that occurred between 1990 and 2012, which amounts to approximately

⁴ Analyzing a large European household panel, Hilber (2014) finds that even conditional on occupant characteristics (including idiosyncratic preferences for homeownership) and location characteristics, a unit in a small apartment building has a roughly 35 percentage points lower probability of being owner occupied than a detached house.

⁵ The US data are based on a 1 percent population sample. The German and Berlin data are based on a subsample of a 1 percent population sample, which is publicly available.

⁶ The number of owner-occupied houses with three or more families is low in Berlin partly because of the heavily subsidized supply of apartments for rent and the restrictive policy on rent levels and rent increases that existed in West Berlin until 1980. Until the end of the German Democratic Republic (GDR), it was almost impossible to privately own an apartment in East Berlin; however, it was possible to own a single-family house.

³ A map of iso-sound pressure zones of 50–55, 55–60, 60–65, 65–67, and greater than 67 decibels was provided by Berlin airports' operating company (Flughafengesellschaft). Ahlfeldt and Maennig (2013) employ a regression-based interpolation to obtain a continuous sound surface within a 100×100 m grid for areas that experience sound pressures above 45 decibels.

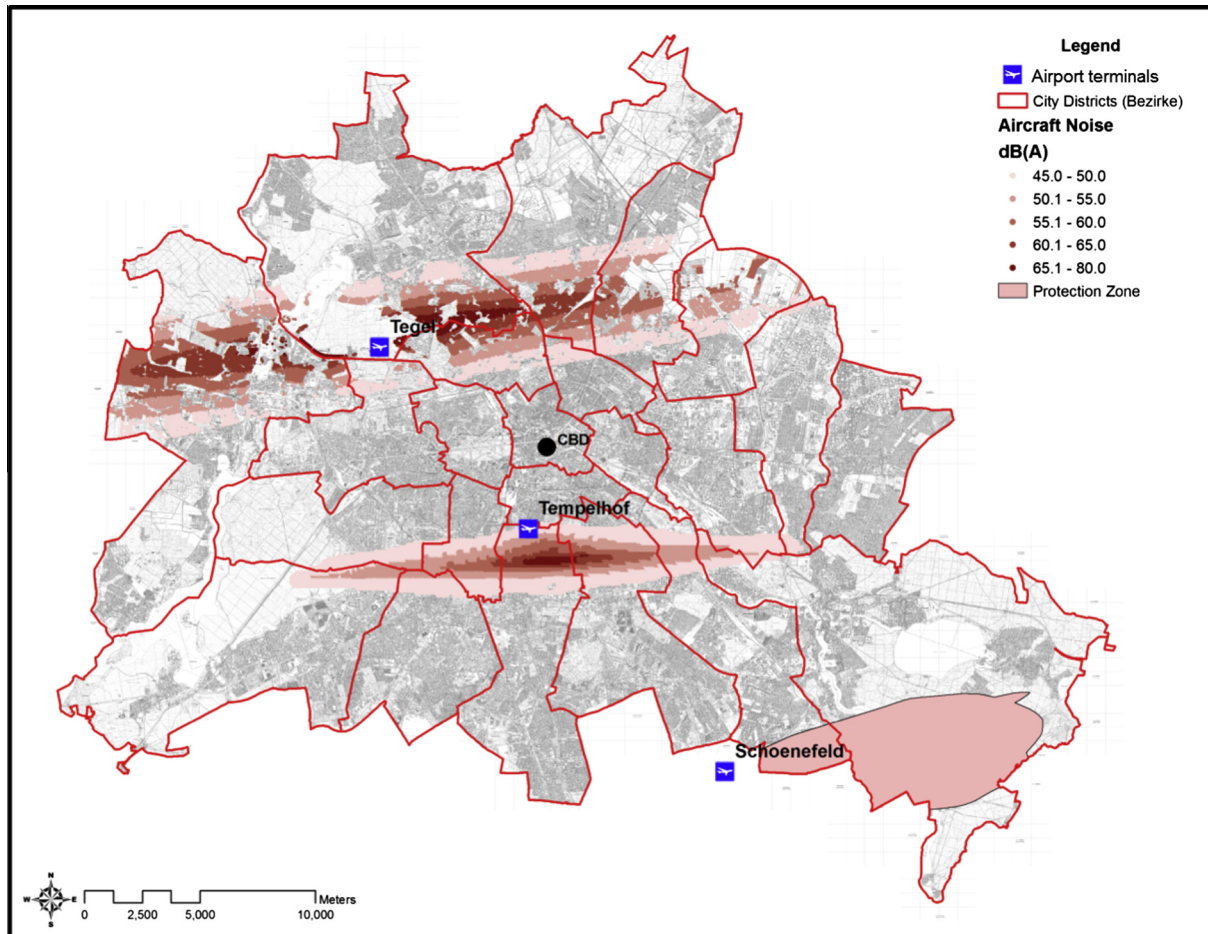


Fig. 1. Airport locations and aircraft noise in Berlin, Germany. *Note:* Authors' illustration using the GIS framework of the Department for Urban Development and the Environment (Senatsverwaltung für Stadtentwicklung Berlin, 2006).

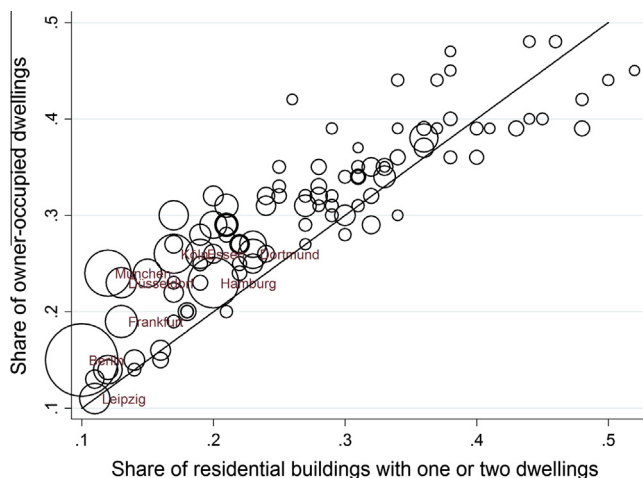


Fig. 2. Owner-occupancy and dwelling structure across the hundred largest German cities in 2011. *Note:* Data are from the regional statistics data base of the German Federal Statistical Office (available at www.destatis.de). The black solid line is the 45-degree line. Marker size is proportionate to city size. The ten largest cities are labeled.

75,000 transactions. The data set includes price, transaction date, location, and a set of variables describing building/plot characteristics. The data are obtained from the Committee of Valuation Experts (Gutachterausschuss Berlin), which is charged with the

statistical analysis of the land registry data for the Department for Urban Development and the Environment. The transactions are geo-referenced by Cartesian coordinates (easting–northing) in projected meter units, which allows them to be integrated into a GIS environment.⁷ The building characteristics include floor space, plot area, surface area, age and age squared, land use, quality of the building stock, location within a block of houses (e.g., a corner lot), and several other amenities, such as basements and elevators. We have also collected a relatively large set of locational variables, including distance to the CBD, the nearest green space, the nearest water body, the nearest metro rail station, the nearest main street, the nearest landmark, the nearest school, and the nearest industrial zone as well as the number of landmarks within 600 m and the street noise level (in decibels).

3. Price signal

Capitalization effects of aircraft noise and airport accessibility are well documented (Ahlfeldt and Maennig, 2013; Boes and Nüesch, 2011; Nelson, 2004). The announcement of the aviation concept on July 4, 1996, implied a substantial change in the distribution of aircraft noise and the access to airport services that was difficult to anticipate. We estimate the property market

⁷ The Department for Urban Development and the Environment uses the Cartesian coordinate system *Soldner*. The origin of the coordinate system is in the south-west of Berlin.

Table 1

Tenure status by dwelling type: Comparison between the US, Germany, and Berlin.

	Dwelling units ≤ 2		Dwelling units > 2		Total
	Total	Percent	Total	Percent	
Tenure status by dwelling type in the US, 2010–2012 ^a					
Owner	958,883	79.86	37,724	15.04	996,607
Renter	214,768	20.14	213,131	84.96	454,899
Total	1,200,651		250,855		1,451,506
	Dwelling units ≤ 2		Dwelling units > 2		Total
	Total	Percent	Total	Percent	
Tenure status by dwelling type in Germany, 2002 ^b					
Owner	10,372	78.68	1943	16.60	12,315
Renter	2632	19.97	9542	81.53	12,174
Subtenant	178	1.35	218	1.86	396
Total	13,182		11,703		24,885
	Dwelling units ≤ 2		Dwelling units > 2		Total
	Total	Percent	Total	Percent	
Tenure status by dwelling type in Berlin, 2002 ^b					
Owner	134	90.54	54	6.42	188
Renter	14	9.46	777	92.39	791
Subtenant	0	0.00	10	1.19	10
Total	148		841		989

Notes: The table reports sample counts based on the sample sizes provided in the American Community Survey and the German Microcensus.

^a US data are based on a 1 percent population sample from the American Community Survey for 2010, 2011, and 2012. The data are accessible via the Integrated Public Use Microdata Series (IPUMS) (available at www.ipums.org; see Ruggles et al., 2010).

^b German data are based on a subsample of a 1 percent population sample from the German Microcensus. The data are publicly accessible (available at www.forschungsdatenzentrum.de).

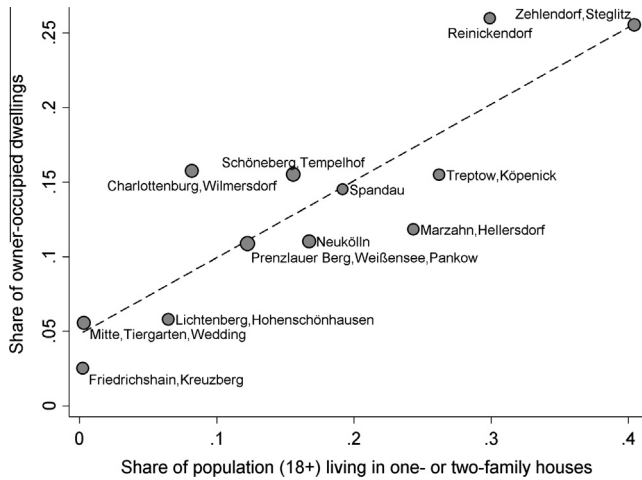


Fig. 3. Estimated homeownership rate vs. actual share of owner-occupied dwellings across Bezirke in Berlin. Note: The dashed line is the linear fit. The share of population living in one- or two-family houses is a population weighted Bezirke level aggregate of our proxy for the homeownership rate (H). The shares of owner-occupied dwellings are as of December 31, 2007 (IBB, 2008).

response to this announcement using a combined hedonic (Rosen, 1974) and difference-in-differences specification:

$$\begin{aligned}
 \log(P_{it}) = & \sum_n \alpha_n^N \text{NOISE}_{in} + \sum_n \alpha_n^D \text{DIST}_{in} + \sum_n \alpha_n^E \text{DISTGT2KM}_{in} \\
 & + \sum_n \beta_n^N (\text{POST96}_t \times \text{NOISE}_{in}) + \sum_n \beta_n^D (\text{POST96}_t \\
 & \times \text{DIST}_{in}) + \sum_n \beta_n^E (\text{POST96}_t \times \text{DISTGT2KM}_{in}) \\
 & + \beta \text{POST96}_t + \gamma \text{TREND}_t + \sum_o b_o X_{oi} + \mu_j + \varepsilon_{it} \quad (1a)
 \end{aligned}$$

where P_{it} is the transaction price of property i at time t . TREND_t is a daily time trend with a zero value at the day before the announcement, X_o is a vector of structural and locational control variables (and b_o are the implicit prices), μ_j is a set of (1201) precinct fixed effects that control for unobserved time-invariant locational characteristics, and ε_{it} is a random zero-mean error term. We capture the airport effects by a set of variables for each of the airports (Tegel, Schoenefeld, and Tempelhof), which we index by n . NOISE_{in} captures the aircraft noise related to airport n at property location i . It is measured as noise level in excess of 45 decibels for the Tegel and Tempelhof airports. Because of data limitations, we use a dummy variable for Schoenefeld, which equals one if property i is within the noise abatement zone and zero otherwise. DIST_{in} is the straight-line distance, in kilometers, to the terminal building of airport n from property location i . To allow for some nonlinearity in the marginal effect of distance, we allow for a spline at the maximum walking distance, which we set to 2 km (Gibbons and Machin, 2005). To empirically accommodate the spline, we add for each airport n an excess distance measure DISTGT2KM_{in} , which takes the value of $\text{DIST}_{in} - 2$ km if DIST_{in} is larger than 2 km and zero otherwise. We differentiate between the period before and the period after the announcement, using a dummy variable POST96_t , which equals one after the announcement and zero otherwise. The interaction terms between POST96_t and the airport variables capture changes in property prices from the pre- to the post-announcement period that are correlated with the airport variables. α , β , and γ are coefficients to be estimated, where higher N , D , and E index noise, distance, and excess distance, respectively.

Table 2 summarizes the estimation results of Eq. (1a). We restrict the presentation of coefficient estimates to the interaction effects between POST96 and the airport variables for brevity. Some, though not all, of the interaction effects between the POST96 identifier and the airport variables are significant, suggesting that the announcement may have had a significant impact on property prices in some areas. From these estimates, it is straightforward

to form an estimate of the announcement effect of the aviation concept (T), which forms the basis of the price signal that we will use in the next stage of the analysis:

$$T_i = \hat{\beta} + \sum_n \hat{\beta}_n^N \text{NOISE}_{in} + \sum_n \hat{\beta}_n^D \text{DIST}_{in} + \sum_n \hat{\beta}_n^E \text{DISTGT2KM}_{in} \quad (1b)$$

where $\hat{\beta}_n^N$, $\hat{\beta}_n^D$, and $\hat{\beta}_n^E$ denote the estimated values of the respective parameters in Eq. (1a). Thus, the announcement effect T_i expresses the effect of the new aviation concept on property prices as a function of noise caused by each of the three airports n as well as the distance to each of the three airports. Essentially, T_i is the composite effect of the changes in implicit hedonic prices of each of the airport-related (pre-announcement) characteristics on property prices at location i .

To test whether T , the joint effect of the individual airport variable effects, has a statistically significant impact on property prices, we replace the individual airport variable $\times$ POST96 interactive terms in Eq. (1a) with an interaction of the POST96 identifier and T in Table 3. Because T is a generated regressor, we bootstrap standard errors (Gawande, 1997; Pagan, 1984).⁸ The estimated coefficient on $\text{POST96} \times T$ in column (1) of Table 3, which is equal to one by construction, is estimated at a relatively high statistical precision.

In Table 2 and column (1) of Table 3, we focus on a time frame ranging from one year before to one year after the announcement day. This short observation period makes it less likely that the estimated announcement effect is driven by long-run spatial adjustments that are accidentally correlated with the airport locations. To empirically test for whether our estimate of the announcement effect T_i is correlated with unobserved spatiotemporal trends, we next add interactions of the time TREND with the POST96 identifier ($\text{TREND} \times \text{POST96}$), the estimated announcement effect T ($\text{TREND} \times T$), and both of these measures ($\text{TREND} \times \text{POST96} \times T$). We experiment with different temporal windows (samples based on years before and after the announcement) using linear and quadratic TREND specifications for larger windows throughout columns (2–5).⁹ With this specification, which integrates elements of difference-in-difference and regression discontinuity designs (Imbens and Lemieux, 2008), we provide a test of whether a significant discrete adjustment occurs at the announcement date conditional on temporal trends that are correlated with T_i . The $\text{POST96} \times T$ interaction effect remains significantly different from zero. The effect is also relatively close to and not statistically different from one. We conclude that it is unlikely that the identified announcement effect T is attributable to unobserved temporal trends that are correlated with the airport variables.

To analyze homevoter and leasevoter effects in the next stage of our analysis, we require a precinct-level measure of a price signal associated with the announcement of the aviation concept. We use the standard transformation (Halvorsen and Palmquist, 1980) to transform the estimated announcement effect T (in log points) into a price signal in percentage terms: $S_j = (\exp(T_j) - 1) \times 100$. We note that T_j is constructed for precinct j according to Eq. (1b) by replacing property i -specific airport variables with the precinct j -specific equivalents. All the distance variables are computed for the geographic centroids of the precincts. Likewise, the Schoenefeld noise variable is coded as a dummy variable that takes the value of one if a precinct centroid is within the noise abatement zone and zero otherwise. The Tegel and Tempelhof noise

variables are expressed as average noise level (in excess decibel) within precinct j . Fig. 4 maps the resulting price signal.

The map reveals positive price responses along the air corridors for Tempelhof, which is intuitive since the announced closure of the airport implied future reductions in the noise disamenity. The mean increase in property prices within the area exposed to aircraft noise (exceeding 45 decibel) is 12 percent. Property prices in the area immediately adjacent to the terminal building (to the north of the airfield) decrease, which is consistent with significant benefits associated with access to airport services.

The area around Schoenefeld where all future aviation activity was announced to be concentrated shows the opposite pattern. The effects are negative within the air corridor but positive adjacent to the terminal building. Property prices fell by 16 percent within the noise abatement zone. With a -0.6 percent price effect per decibel from the literature (Nelson, 2004), the change in aircraft noise implied by our estimates corresponds to 25 decibels, roughly consistent with the increase in noise that could be expected.¹⁰ The effects for Tegel, the more remote of the two airports that were announced to close, are generally smaller and restricted to a narrower area around the airfield.

4. Voting patterns

4.1. Airport effects

Fig. 5 maps the support for the new aviation concept based on the share of NO votes on the referendum on the continued operation of Tempelhof. The east–west heterogeneity evident from the map is striking. The (unconditional) mean approval rate for the new concept is more than twice as high within former East Berlin as within former West Berlin (65 percent vs. 30 percent), which is another impressive example of the economic legacy of the Berlin Wall (Ahlfeldt et al., forthcoming). Similarly impressive is the clearly reduced support for the new concept within the Schoenefeld noise zone (27 percent), whereas the increases in the approval rates for the new concept around Tegel and Tempelhof are visible but less obvious. The effects in the immediate vicinity of the aircraft noise zones are suggestive of NIMBYism and the desire to shift (Tempelhof and Tegel) or keep (Schoenefeld) airport activity away from the voters' own neighborhoods.

To analyze whether support in the referendum depends on the spatial distribution of aircraft noise and airport accessibility, we regress the share of NO votes against the precinct-level equivalents of the airport variables that are used in Section 3 NOISE_{jn} , DIST_{jn} , and DISTGT2KM_{jn} . Column (1) of Table 4 substantiates our interpretations of Fig. 5. An increase in aircraft noise by one excess decibel triggers an increase in the proportion of NO votes by approximately 0.9 percentage points for precincts affected by Tempelhof noise and by approximately 0.2 percentage points in the case of Tegel. Given the maxima of the observed noise levels for the precinct average of approximately 14 excess decibels for Tempelhof and 30 excess decibels for Tegel, these estimates indicate a substantial increase in support for the new airport concept of up to 13 (Tempelhof) and 7 (Tegel) percentage points. As expected, the aircraft noise effect is weaker for Tegel than Tempelhof because the former is less directly related to the referendum, which directly addressed the closure of Tempelhof. The support for the new airport concept within the Schoenefeld noise zone is significantly reduced (21 percentage points lower).

⁸ The estimation equation thus collapses to $\log(P_{it}) = \theta_0 T_i + \theta(T_i \times \text{POST96}_i) + \beta \text{POST96}_i + \sum_o b_o X_{oi} + \mu_j + \varepsilon_{it}$.

⁹ The estimation equation takes the following form: $\log(P_{it}) = \theta_0 T_i + \theta(T_i \times \text{POST96}_i) + \beta \text{POST96}_i + \sum_m \gamma_m \text{TREND}_t^m + \sum_m \gamma_m^{\text{POST96}} (\text{POST96} \times \text{TREND}_t^m) + \sum_m \delta_m^{\text{POST96}} (T_i \times \text{POST96} \times \text{TREND}_t^m) + \sum_o b_o X_{oi} + \mu_j + \varepsilon_{it}$, where $M = 1$ implies a linear trend specification and $M = 2$ implies a quadratic trend specification.

¹⁰ The highest noise levels measured at Schoenefeld airport in 2006 were slightly below 60 decibels (Flughafen Berlin Schoenefeld, 2008). The highest noise level recorded for Tegel, Berlin's busiest airport, is 80 decibels in our data (see Fig. 1).

Table 2

Capitalization effects of the 1996 announcement: Difference-in-differences estimates by airport.

Airport n	(1) Log of property price (P)					
	Tempelhof		Tegel		Schoenefeld	
$POST96 \times NOISE_n$	0.015 [*]	(0.009)	–0.001	(0.003)	–0.186	(0.132)
$POST96 \times DIST_n$	–0.287	(0.414)	0.123	(0.089)	–2.929 [*]	(1.670)
$POST96 \times DISTGT2KM_n$	–0.119	(0.091)	0.280	(0.415)	2.923 [*]	(1.671)
Hedonic controls	YES					
Precinct fixed effects	YES					
Airport variables	YES					
POST96	YES					
N	6796					
R^2	0.785					

Notes: OLS estimates. The dependent variable is the log of property price (P). $POST96$ is a dummy variable, which equals one after the 1996 announcement and zero otherwise. $NOISE_n$ is the noise level in excess of 45 decibels associated with airport n in case of Tempelhof and Tegel and a 0,1 dummy variable, which is one for locations within the noise abatement zone and zero otherwise, in the case of Schoenefeld. $DIST_n$ is the straight-line distance to the terminal building of airport n in km. $DISTGT2KM_n$ is equal to $DIST_n - 2km$ if $DIST_n$ is larger than 2 km and zero otherwise. Hedonic controls include lot size and lot size squared, number of floors, floor area ratio and floor area ratio squared, building age and building age squared, condition of the building (dummies for good and bad), dummies for penthouse, elevator, basement, being vacant, and being located in an urban renewal area; distance to the central business district, the nearest green spaces, the nearest water body, the nearest metro rail station, the nearest main street, the nearest landmark, the nearest school, and the nearest industrial zone; the number of landmarks within 600 m; and the street noise level (decibel). Airport variables are (non-interacted) $NOISE_n$, $DIST_n$, $DISTGT2KM_n$ for each airport n . The observation period ranges from one year before to one year after the 1996 announcement. Standard errors in parentheses are clustered on precincts.

* $p < 0.10$.** $p < 0.05$.*** $p < 0.01$.**Table 3**

Capitalization effects of the 1996 announcement: Difference-in-differences estimates conditional on time trends.

	(1) Log(P)	(2) Log(P)	(3) Log(P)	(4) Log(P)	(5) Log(P)
$POST96 \times T$	1.000*** (0.415)	1.267 [*] (0.739)	1.196** (0.557)	0.938** (0.447)	0.813** (0.342)
Hedonic controls	YES	YES	YES	YES	YES
Precinct fixed effects	YES	YES	YES	YES	YES
Treatment (T)	YES	YES	YES	YES	YES
POST96	YES	YES	YES	YES	YES
Period in years before and after announcement included	1	1	2	5	10
Linear trend effects	NO	YES	YES	NO	NO
Quadratic trend effects	NO	NO	NO	YES	YES
N	6,796	6,796	13,678	34,241	57,913
R^2	0.785	0.785	0.742	0.735	0.726

Notes: OLS estimates. The dependent variable is the log of property price (P) in all models. T is the announcement effect of the aviation concept, computed according to Eq. (1b) and the estimates shown in Table 2. $POST96$ is a dummy variable, which equals one after the 1996 announcement and zero otherwise. Hedonic controls are the same as in Table 2. Linear and quadratic trend effects are linear or quadratic monthly trends, trends interacted with the $POST96$ dummy, trends interacted with the joint treatment T , and trends interacted with T and $POST96$. The observation period in column (5) begins in 1990 (January 1) because of data availability. Standard errors in parentheses are clustered on precincts and bootstrapped with 500 iterations.

* $p < 0.10$.** $p < 0.05$.*** $p < 0.01$.

These results indicate that aircraft noise was generally perceived to be a disamenity by the local population.

The distance effects satisfy the significance criteria for Tempelhof only. The share of NO votes decreases by 7.51 percentage points per km within the first 2 km. Beyond this threshold, the marginal effect is reduced to $(7.51 - 4.82) = 2.71$. This result suggests that voters living close to the terminal, but outside the major noise corridor, supported the initiative. This finding is at odds with the pattern of price adjustments shown in Fig. 4. However, the evidence is reconciled once we control for preferences that are correlated with observable socioeconomic characteristics of the local population—see column (2) of Table 4. Conditional on these controls, support for the new aviation concept first increases in distance to the Tempelhof terminal (beyond 2 km) and then decreases. The large support in these areas seems largely attributable to the high share of Green and low share of conservative party

voters. This is consistent with the clear position the parties took in favor of and against the referendum.

Column (3) uses turnout as a dependent variable. Turnout is higher in areas where Tempelhof aircraft noise would decrease, as well as where Schoenefeld aircraft noise would increase, with the new aviation concept. Since participation was higher where either benefits or costs had to be associated with the new aviation concept, it seems unlikely that many voters remained absent strategically. This notion is consistent with a relatively high turnout compared to other referenda in Berlin.¹¹ It is also consistent with

¹¹ The turnout in the 2008 Tempelhof Airport referendum was 36.1 percent, compared to 29.2 percent (2009, “ethics/religion” as a compulsory school subject), 27.5 percent (2011, publication of the contracts of the privatization of Berlin Water Supply), and 29.1 percent (2013, remunicipalization of the energy supply) in the three other referenda that have ever been held in Berlin.

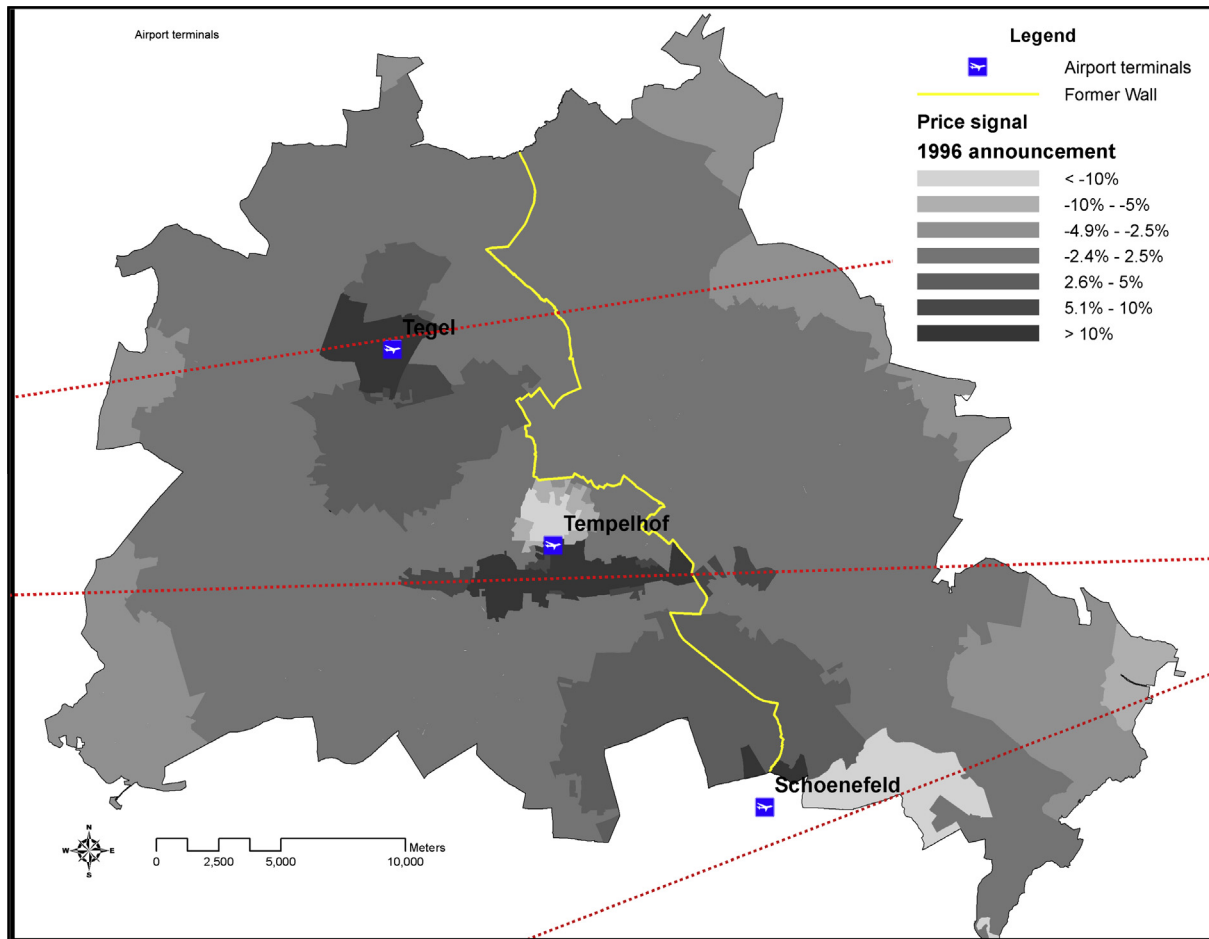


Fig. 4. Price signal: Estimated price effects of the 1996 announcement of the new Berlin aviation concept. *Note:* Authors' illustration based on the GIS framework of the Department for Urban Development and the Environment (Senatsverwaltung für Stadtentwicklung Berlin, 2006). The price signal $S_j = (\exp(T_j) - 1) \times 100$ is constructed for precincts based on Eq. (1b) and the parameter estimates from Table 2. The dotted lines are extensions of the airport runways that roughly approximate the flight path of approaching and departing aircraft.

a quorum that requires 25 percent of YES votes of the total electorate rather than a minimum turnout.

4.2. Homevoter vs. leasevoter effects

In analyzing the voting pattern for the 2008 Tempelhof referendum, we assume that voters who participate in a public referendum support the alternative that maximizes their expected utility. A price signal that is associated with an initiative in question is a critical component in this reasoning. Approval of an initiative implies that the anticipated capitalization effects remain persistent or are strengthened if the announcement effects are initially discounted on uncertainty. Rejection of the initiative instead implies that the anticipated capitalization effects will be reversed.

To test for homevoter vs. leasevoter effects, we relate the support for the new aviation concept in the referendum to our estimated price signal and our approximation of the homeownership rate. The empirical specification that we use shares many similarities with that of Dehring et al. (2008). The main difference between our specification and theirs is that we concentrate on the interaction effect between homeownership rate H and price signal S , which allows for an explicit differentiation in the response to similar price signals between homevoters and leasevoters:

$$\begin{aligned}
 PRCNTNO_j = & \alpha + \Omega H_j + \phi S_j + \psi(H_j \times S_j) + \sum_k A_k Z_{kj} \\
 & + \sum_n \omega_n^N NOISE_{jn} + \sum_n \omega_n^D DIST_{jn} \\
 & + \sum_n \omega_n^E DISTGT2KM_{jn} + \epsilon_j
 \end{aligned} \quad (2)$$

The support for the new airport project is expressed as the percentage of NO votes in total votes for voting precinct j ($PRCNTNO$). Since voters' attitudes in public referenda are likely shaped by needs (the economic situation) and sympathies (their political orientation) (Brunner et al., 2011), we add socioeconomic covariates Z_k , which are introduced in Table 4.

The variables of primary interest are a proxy for the precinct-level homeownership rate H (in percent), discussed in more detail in the data section, and the price signal $S_j = (\exp(T_j) - 1) \times 100$, plotted in Fig. 4. Because S_j is a generated regressor, we bootstrap the standard errors when estimating Eq. (2) (Gawande, 1997; Pagan, 1984).

The price signal is the market capitalization of the expected changes in service flow associated with the new aviation concept and the amenity changes it brings. The interpretation of the composite effect of the price signal, the homeownership rate, and their interaction ($\Omega H_j + \phi S_j + \psi(H_j \times S_j)$) on the voting behavior therefore depends on whether the airport variables

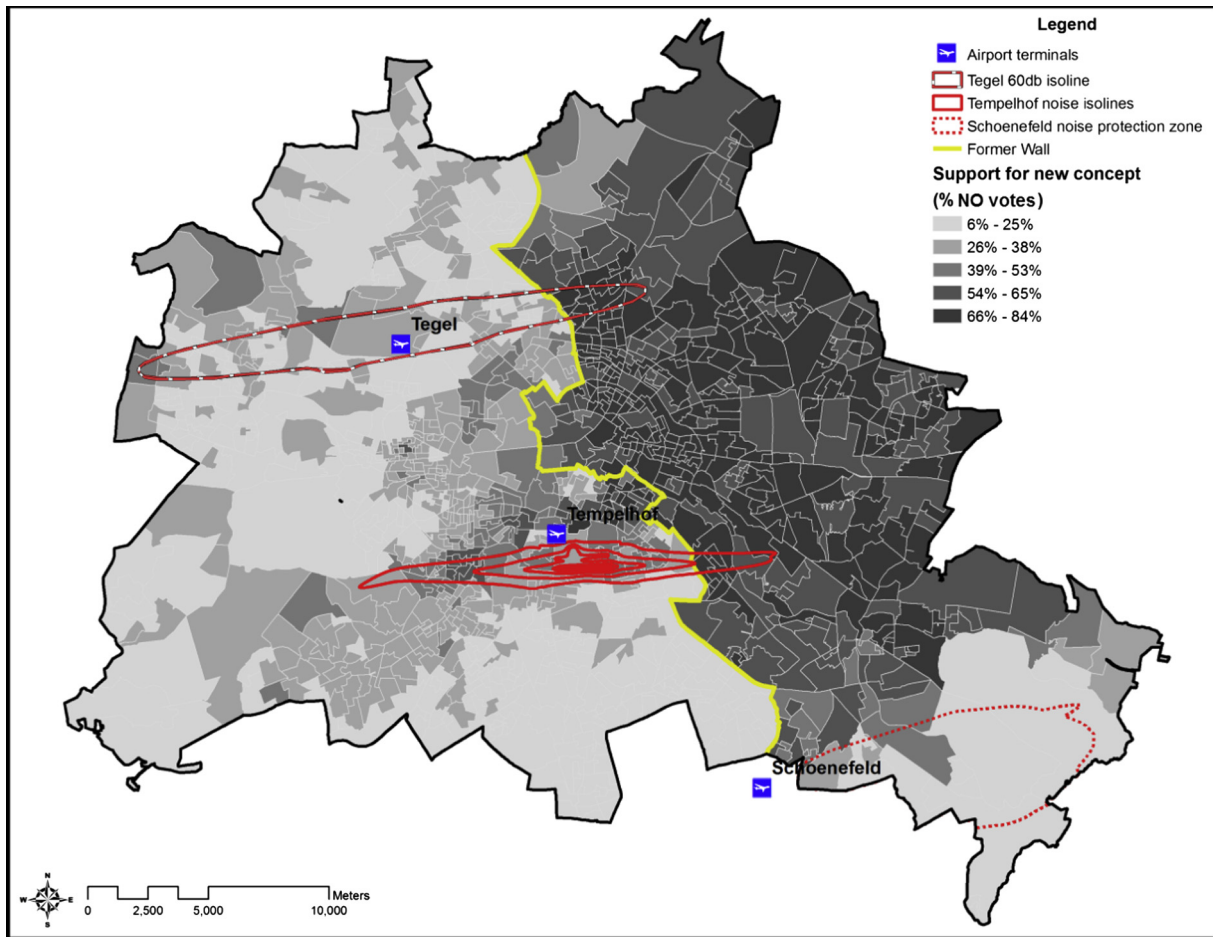


Fig. 5. Support for the new aviation concept: Share of NO votes in the 2008 referendum on the continued operation of Tempelhof. *Note:* Authors' illustration based on the GIS framework of the Department for Urban Development and the Environment (Senatsverwaltung für Stadtentwicklung Berlin, 2006).

$NOISE_{jn}$, $DIST_{jn}$, and $DISTGT2KM_{jn}$ (discussed in detail in the context of Eq. (1a)) are included in the model.

In line with Dehring et al. (2008), we argue that if the airport variables are excluded from the model, the effect of the price signal and its interaction with the homeownership rate on the voting behavior capture the composite of a *consumption effect* and a *wealth effect*. The consumption effect originates from the expected change in service flow consumption caused by the new aviation concept and the effects on the utility of voters who stay in their properties. The wealth effect originates from the capitalization effects caused by the new aviation concept. For homeowners, the wealth effect arises from the expected change in housing value that can be realized by selling or renting out a property after a household moves out. For renters, the wealth effect arises from a change in the expected net present value of future rental payments to the landlord over the time a household stays in the property. If the airport variables are added to the model, they control for the consumption effect so that the effect of the price signal and its interaction with the homeownership on the voting behavior captures the wealth effect exclusively.

Positive price signals should be associated with support among homevoters because the consumption effect and the wealth effect work in the same direction. Likewise, homevoter opposition to negative price signals is expected. For leasevoters, positive amenity changes imply a positive consumption effect and a negative wealth effect due to expected increases in rent. Likewise, a negative consumption effect due to a negative amenity change is mitigated by a positive wealth effect due to an expected reduction in the

net present value of future rental payments. This opposite direction of the wealth effect provides a source of heterogeneity in the voting behavior of homevoters and leasevoters. Positive price signals should be supported and negative signals opposed more strongly by homevoters than leasevoters. Because the homeownership rate–price signal interaction effect captures the wealth effect regardless of whether the airport control variables are included or not, the interaction effect must be positive in either case to confirm the homevoter hypothesis.

Table 5 presents various empirical estimates of this interaction. We first estimate a model that, apart from a dummy for East Berlin, only includes the homeownership rate, the price signal, and the interaction of both variables (1) and then subsequently add socio-economic controls (2) as well as a full set of interaction terms between the controls and the homeownership rate (3). These interactive terms make it less likely that the homeownership rate–price signal interaction term captures an interaction effect of tenure and some attributes of the population that are spatially correlated with the price signal but not causally related to the changes induced by the new aviation concept. As expected, the interaction effect of interest is positive and statistically significant in all model variants.

Next, we add the airport noise and accessibility variables to the model in columns (4) and (5). With these variables, we control for the consumption effect so that the homeownership rate and price signal interaction captures the wealth effect exclusively. The interaction effect decreases relative to the comparable models that exclude the airport variables (2 and 3), but remains significant. Column (6) replicates column (5) by weighting the observations

Table 4

Voting behavior: Baseline models.

	(1) NO votes [percent]		(2) NO votes [percent]		(3) Turnout [percent]	
(Former) East Berlin [dummy]	36.35***	(0.68)	25.77***	(1.37)	−8.07***	(0.76)
Tempelhof <i>NOISE</i> [excess decibels]	0.87***	(0.26)	0.76***	(0.20)	0.65***	(0.08)
Tegel <i>NOISE</i> [excess decibels]	0.22***	(0.05)	0.27***	(0.04)	−0.02	(0.03)
Schoenefeld <i>NOISE</i> zone [dummy]	−21.31***	(5.48)	−27.63***	(5.86)	6.46**	(2.79)
<i>DIST</i> ance to Tegel [km]	0.62	(2.77)	3.82*	(2.14)	−3.00	(1.94)
<i>DIST</i> ance to Tempelhof [km]	−7.51***	(2.18)	3.11***	(1.06)	−2.67***	(0.54)
<i>DIST</i> ance to Schoenefeld [km]	−12.62	(12.60)	14.55	(13.77)	16.45***	(4.19)
<i>DISTGT2KM</i> :Tegel [km]	0.90	(2.80)	−3.44	(2.17)	2.81	(1.95)
<i>DISTGT2KM</i> :Tempelhof [km]	4.82**	(2.25)	−4.20***	(1.09)	2.78***	(0.57)
<i>DISTGT2KM</i> :Schoenefeld [km]	14.04	(12.63)	−14.09	(13.78)	−16.76***	(4.21)
Homeownership rate <i>H</i> [percent]			0.06***	(0.01)	0.05***	(0.01)
Share conservative parties [percent]			−0.81***	(0.05)	0.12***	(0.03)
Share green party [percent]			0.73***	(0.05)	0.16***	(0.03)
Share age < 18 [percent]			0.63***	(0.09)	0.02	(0.05)
Share age 18–27 [percent]			0.85***	(0.12)	−0.16**	(0.07)
Share age 45–55 [percent]			0.71***	(0.08)	−0.10**	(0.04)
Share age > 55 [percent]			0.54***	(0.05)	0.03	(0.03)
Purchasing power [1000 Euro/capita]			0.25	(0.21)	−0.34***	(0.13)
Share non-Germans [percent]			−0.18***	(0.05)	−0.25***	(0.02)
Share unemployed [percent]			−0.06	(0.10)	−0.52***	(0.06)
Constant	63.95**	(25.54)	−4.53	(27.86)	10.40	(9.14)
<i>N</i>	1201		1201		1201	
<i>R</i> ²	0.82		0.92		0.80	

Notes: OLS estimates. NO votes is the share of NO votes among voters who voted in person. NOISE is measured in excess of 45 decibels. DISTance is measured as a straight line to the terminal building. DISTGT2KM is equal to $DIST - 2$ km if DISTance to the respective airport is larger than 2 km and zero otherwise. Robust standard errors are in parentheses, and units in brackets.

* $p < 0.10$.** $p < 0.05$.*** $p < 0.01$.**Table 5**

Support for the new airport concept: Interaction effects of homeownership rate and the 1996 price signal.

	(1) NO votes [percent] OLS	(2) NO votes [percent] OLS	(3) NO votes [percent] OLS	(4) NO votes [percent] OLS	(5) NO votes [percent] OLS	(6) NO votes [percent] WLS
Homeownership rate <i>H</i> [percent]	–0.115*** (0.011)	0.022* (0.014)	0.015 (0.021)	0.056*** (0.011)	0.023 (0.018)	0.025 (0.017)
1996 Price signal <i>S</i> [percent]	–0.300** (0.136)	–0.201 (0.173)	–0.140 (0.148)	–0.424 (2.305)	–0.380 (2.655)	–0.384* (0.209)
Price signal <i>S</i> × Homeownership rate <i>H</i>	0.012*** (0.005)	0.011** (0.005)	0.009** (0.004)	0.009** (0.004)	0.008** (0.004)	0.006** (0.003)
Controls	NO	YES	YES	YES	YES	YES
Controls × <i>H</i>	NO	NO	YES	NO	YES	YES
Airport variables	NO	NO	NO	YES	YES	YES
Weighted by electorate	NO	NO	NO	NO	NO	YES
<i>N</i>	1201	1201	1201	1201	1201	1201
<i>R</i> ²	0.725	0.889	0.900	0.917	0.924	0.932

Notes: The dependent variable is the share of NO votes among voters who voted in person in all models. All models include a dummy variable to differentiate between former East and West Berlin. Controls include the share of green and conservative party votes, several age groups, and the non-German population; disposable household income; and the unemployment rate. Controls × *H* are the same variables interacted with the homeownership rate. All the control variables have a mean of zero. Airport variables are the aircraft noise and distance (with a spline at 2 km) variables for all three airports *n* introduced in Eq. (1) ($NOISE_n, DIST_n, DISTGT2KM_n$). Column (6) estimates are weighted by the precinct electorate. Standard errors in parentheses are bootstrapped with 500 iterations in columns (1–5).

* $p < 0.10$.** $p < 0.05$.*** $p < 0.01$.

by the precinct electorate. The effect further decreases but, once more, is positive and significant. Thus, the result supports the homevoter hypothesis.

The homevoter hypothesis lends itself to the derivation of some more-explicit predictions. Because consumption and wealth effects work in the same direction for homevoters, conditioning out consumption effects should reduce support in areas with positive price signals and high homeownership. Similarly, opposition to negative price signals should decrease in high-homeownership areas. Because consumption and wealth effects work in opposite

directions for leasevoters, controlling for the consumption effect should increase opposition in areas with positive price signals and low homeownership rates and increase support in areas with negative price signals and low homeownership rates.

To allow for an intuitive evaluation of these predictions, we visualize the composite effect of the homeownership rate, the price signal, and their interaction ($\Omega H_j + \phi S_j + \psi(H_j \times S_j)$) on the voting outcome on a quasi-spatial 3D surface in Fig. 6. We present two surfaces, one based on a model without airport variables and another based on a mode with airport variables. Thus, the former

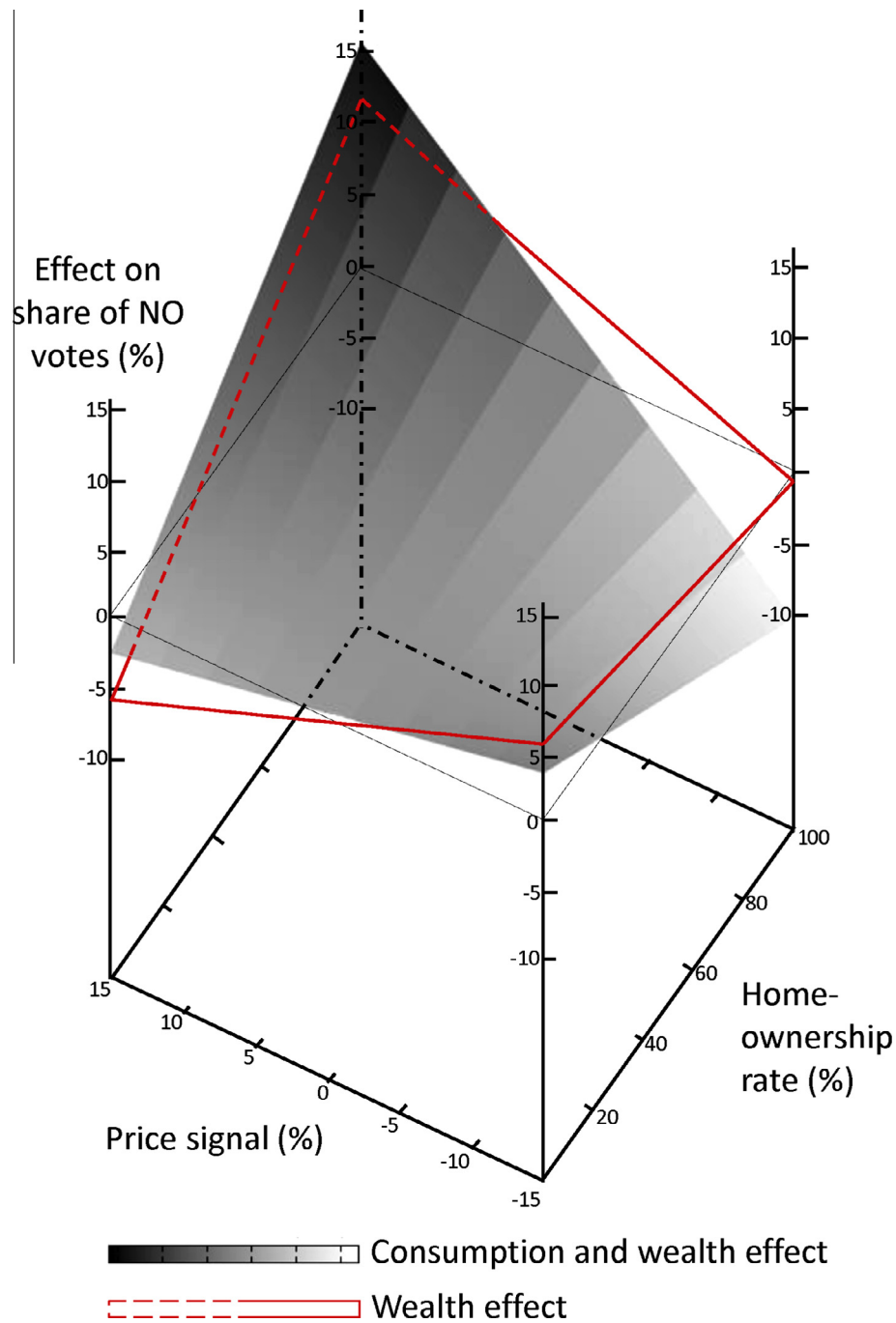


Fig. 6. Support for the new airport concept: Interaction effects of homeownership rate and the 1996 price signal. *Note:* The surfaces represent the joint effects of homeownership rate H and the price signal S on the percentage share of NO votes $(\hat{\Omega}_j^H + \hat{\phi}_j^S + \hat{\psi}(H_j \times S_j))$, estimated according to Eq. (2). The shaded surface is based on the model in Table 5, column (2), which excludes airport-related control variables. The surface, thus, illustrates the composite consumption and wealth effect of the new aviation concept on voters' decisions. The outlined surface is based on the model in Table 5, column (4), which includes airport-related control variables. The surface, thus, illustrates the pure wealth effect. Controlling for the consumption effect tends to reduce homeowner effects (at high homeownership rates) and to increase leaseholder effects (at low homeownership rates).

captures the composite consumption and wealth effect, based on Table 5, column (2), whereas the latter exclusively reflects the wealth effect, based on Table 6, column (4). We constrain the surfaces to a range of ± 15 percent of the price signal since 98 percent of our precincts fall within this interval.

We first consider the composite consumption and wealth effect (Fig. 6, shaded surface). As expected, we find significantly greater support for the new airport concept in areas with both high homeownership rates and positive price signals (in the northwestern

quadrant). Here, the share of NO votes, *ceteris paribus*, increases by approximately 15.7 percentage points compared to a baseline precinct with a zero homeownership rate and a zero price signal. A negative price signal of -15 percent in such a homeowner neighborhood reduces the share of NO votes by 11.3 percentage points. In pure renter precincts, the effects with similar price signals point to the opposite directions, although the effects are much more moderate (-3 percentage points at a $+15$ percent price signal, and vice versa). The wealth effect, thus, overcompensates for the

Table 6

Interaction effects of homeownership rate and the 1996 price signal on turnout in the 2008 Tempelhof referendum and falsification via placebo referendum.

	(1) Turnout [percent] REF2008	(2) Turnout [percent] REF2008	(3) Turnout [percent] REF2008	(4) NO votes [percent] PLACEBO ^a	(5) NO votes [percent] PLACEBO ^a	(6) NO votes [percent] PLACEBO ^a
Homeownership rate H [percent]	0.126*** (0.005)	0.048*** (0.006)	0.029** (0.013)	−0.118*** (0.008)	0.063*** (0.010)	−0.003 (0.014)
2007 Price signal [percent]	0.117* (0.065)	−0.126 (0.941)	−0.119 (1.009)	−0.052 (0.119)	−0.197 (2.167)	−0.161 (1.784)
Price signal $S \times$ Homeownership rate H	−0.003* (0.002)	−0.002 (0.003)	−0.002 (0.003)	−0.002 (0.002)	0.000 (0.003)	0.001 (0.003)
Controls	NO	YES	YES	NO	YES	YES
Controls $\times H$	NO	NO	YES	NO	NO	YES
Airport variables	NO	YES	YES	NO	YES	YES
N	1201	1201	1201	1201	1201	1201
R^2	0.575	0.799	0.806	0.754	0.933	0.940

Notes: OLS estimates. NO votes is the share of NO votes among voters who voted in person. REF2008 denotes the 2008 referendum on the continued operation of Tempelhof airport. PLACEBO denotes the 2009 referendum on whether ethics should become a compulsory school subject. All models include a dummy variable to differentiate between former East and West Berlin. Controls, Controls $\times H$ and Airport effects are defined in the notes to Table 5. In all, 27 precincts in the PLACEBO election were affected by boundary changes. These precincts could be mapped to 6 meta-precincts for which the election outcome was available according to the new definition. The affected precincts were assigned the mean outcome of the meta-precincts. Standard errors in parentheses are bootstrapped with 500 iterations.

^a As with REF2008, the voting data for the PLACEBO referendum in 2009 was provided by the Statistical Office for Berlin-Brandenburg.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

consumption effect. This status quo preference can be rationalized by heterogeneous preferences and high cost of relocation to other neighborhoods. Incumbent renters fear that they may be displaced by new residents with stronger preferences for environmental quality.

When we control for the consumption effect (Fig. 6, outlined surface), the positive homevoter effect in high price signal precincts is reduced, as expected, but remains remarkably large (about 12.5 percentage points at $H = 100$ and $S = 15$). This large wealth effect suggests that expected capital gains are a relevant motive for homeowners to support positive amenity changes. In contrast, in homevoter precincts with negative price signals, we find virtually no wealth effect. Homevoters do not seem to consider the option to sell their property (at a lower price) to a buyer with lower preferences for a calm neighborhood and to relocate to a (smaller) property in another (potentially quieter) neighborhood. Controlling for the consumption effect, the leasevoter effects increase as expected. At a 15 percent price signal and a zero homeownership rate, the wealth effect reduces the share of NO votes by 6 percentage points, which is about twice as much as the consumption and wealth effects combined.

Our results confirm the homevoter hypothesis: homevoters strongly support initiatives that are expected to increase the amenity value of their area. Our results also add a new insight to the related empirical literature, showing that leasevoters are more indifferent to amenity changes than homevoters because the landlords ultimately accumulate the benefits of such changes. Another notable finding is that when we control for the consumption effect and focus on the wealth effect, the leasevoter opposition induced by positive price signals remains relatively small compared to the homevoter support for similar price signals.

One explanation for this relatively moderate leasevoter effect is the strong rental regulation in Germany. The degree to which landlords can adjust the rents is limited as long as their renters do not move out. The protection of renters is particularly strong because lease contracts usually do not have a finite duration and because the landlord can unilaterally cancel a contract only under special circumstances. Because sitting renters are not fully exposed to increases in market rent, their incentives to oppose the capitalization effects of positive amenity changes are reduced. A complementary explanation is that renters are relatively mobile, unlike homeowners in Germany who face large closing costs associated

with the purchase of real estate. Since renters can respond to undesirable increases in rents caused by an amenity change by moving to another neighborhood, their incentives to oppose initiatives that induce positive capitalization effects are mitigated.

5. Complementary evidence and falsification

Homeowners tend to invest more in social capital (DiPasquale and Glaeser, 1999) and tend to have higher voter participation (Brunner et al., 2001). The large homevoter effects we find could be the result of a combination of homeowners being more likely to vote in favor of an initiative that brings positive amenity change and a particularly high propensity to cast a vote if the expected gain is large. We replicate selected models of Table 5 in Table 6, columns (1–3), with turnout as the dependent variable. The homeownership rate has a significantly positive effect on turnout. The interaction effect between the price signal and the homeownership rate, however, is either insignificant or even negative. Higher voter participation among homevoters is thus unlikely to cause strong homevoter effects.

The main empirical concern with Eq. (2) is arguably that homeowners with a particular unobserved lifestyle orientation could live in areas with similar price signals and vote differently from other homeowners for reasons that are unrelated to the aviation concept. One way to address this concern is to analyze a referendum that is unrelated to the initiative in question using an otherwise similar specification. In Table 6, columns (4–6), we analyze the outcome of the 2009 public referendum on whether ethics should become a compulsory school subject in Berlin. This was the first referendum after the Tempelhof poll. We do not find a significant homeownership–price signal interaction effect in any of the considered placebo models, which increases our confidence that the interaction effect in the baseline models relates to the phenomenon that we are interested in.

A correlation between unobserved voter characteristics and our estimated price signal appears like a potential concern because our estimated price signal evidently shows a high degree of spatial autocorrelation and unobserved homeowner characteristics are also likely correlated in space. To further address the concern that the interaction effect between homeownership rate and price signal might be an artifact of an omitted variable problem, we conduct a second falsification exercise. We consider a series of

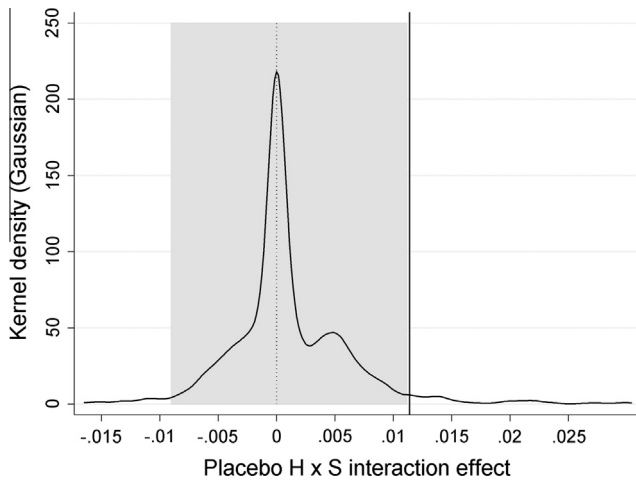


Fig. 7. Falsification: Interaction effects of homeownership rate and the 1996 price signal for placebo airport locations. *Note:* The figure shows the distribution (Gaussian kernel density) of 1000 placebo point estimates of the homeownership–price signal interaction effect in Eq. (2), excluding the airport variables. Except for using a placebo price signal, each placebo test is identical to Table 5, column (2). For each placebo test, we first generate placebo airport locations based on random draws from a uniform distribution of Cartesian coordinates that reflects the spatial extent of Berlin. A price signal is then constructed based on placebo airport variables generated for the placebo airport locations and the parameter estimates from Table 2. The vast majority of placebo tests yield homeownership–price signal ($H \times S$) interaction effects that are close to zero. Our benchmark point estimate (solid vertical line) is about two standard deviations (shaded area) from the mean of the distribution of placebo effects in the right tail and, thus, is unlikely to result from an accidental correlation with unobserved voter characteristics.

randomly generated placebo price signals with a similar spatial structure to ensure that they are similarly likely to capture unobserved characteristics that are correlated in space. We use these placebo price signals in specifications that are otherwise identical to our benchmark models. The distribution of the estimated placebo effects will be informative of how likely it is that an accidental correlation between unobserved voter characteristics and an estimated price signal leads to homeownership–price–signal interaction effects of the magnitude estimated in our benchmark models.

We begin each placebo test by drawing locations for all three airports from a random distribution of Cartesian coordinates that reflect the spatial extent of Berlin. For each placebo airport location, distance variables ($DIST_n$, $DISTGT2KM_n$) are computed for all precincts in the same way as for the actual airport locations. To create noise variables ($NOISE_n$) for the placebo airport locations, we first compute the mean noise levels within 1×1 km grid cells around each of the actual airport locations. For each airport, we construct a grid that consists of 20 1-km steps to the east and west and 5 1-km steps to the north and south, resulting in a total of 20×2 (east/west) $\times 5 \times 2$ (north/south) $\times 3$ (airports) = 1200 grid cells. The size of the airport grids are chosen to ensure that they include the relevant noise information but do not overlap. The mean noise levels in these grid cells are then shifted to precincts according to their relative position in grids that are analogously created for the placebo airport locations. Once airport noise and distance variables are created for the placebo airport locations, it is possible to construct a placebo price signal by using the parameter estimates from Table 2. We use this procedure to construct 1000 placebo price signals, which we then use in 2×1000 estimations that are otherwise identical to those in Table 5, columns (2) (excluding airport variables) and (4) (including airport variables).

The distribution of the homeownership–price signal ($H \times S$) interaction effects across the placebo tests that exclude airport variables is plotted in Fig. 7. The vast majority of placebo tests yield homeownership–price signal ($H \times S$) interaction effects that are

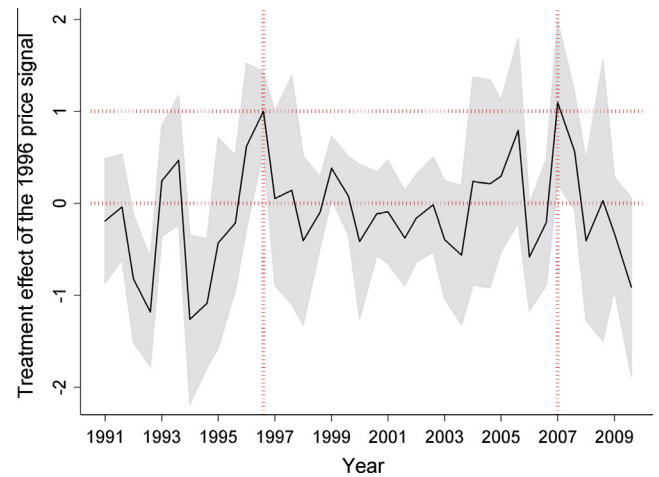


Fig. 8. Hypothetical announcement dates: Difference-in-differences treatment effects of the 1996 price signal on property prices. *Note:* The figure illustrates difference-in-differences type estimates of the effect the 1996 price signal (the treatment measure) on property prices at hypothetical announcement dates (biannual from January 1, 1991 to July 1, 2009), as discussed in footnote 12. The solid line connects the point estimates. The gray dashed area represents the 95 percent confidence interval. Before the 1996 announcement (left dotted vertical line), the 1996 price signal is, if anything, negatively correlated with price trends. Volatility in the estimated parameters is high. At the 1996 announcement, the estimated parameter is one by construction. After that, the 1996 price signal yields a significant treatment effect on property prices only at the hypothetical announcement date, January 1, 2007 (right dotted vertical line), which coincides with the final decision by the Federal Administrative Court of Germany as the court of last instance.

close to zero. Our benchmark point estimate (solid vertical line) is about two standard deviation lengths (gray shaded area) from the mean of the distribution of the placebo effects in the right tail. The distribution of placebo tests that include airport variables, presented in the Appendix, is very similar. We conclude that the benchmark interaction effects reported in Table 5 are unlikely to result from an accidental correlation with unobserved voter characteristics.

Our final alteration of the benchmark model considers other potential price signals. With the 1996 announcement, the arguably critical information regarding the future of aviation in Berlin was released. It is possible, however, that subsequent to the initial announcement, additional events affected the voters' expectations regarding whether or when the new concept would be implemented. Unlike for the initial announcement, announcement dates for such events are difficult to determine given the relatively long, controversial, and complex debate that eventually led to the Tempelhof referendum. To search for spatial adjustments in property prices that are correlated with the 1996 announcement effect, we use model (1) from Table 3. The model is replicated for a series of biannual (January 1 and July 1) hypothetical announcement dates ranging from 1991 to 2009.¹² We expect to find positive treatment estimates if important information was released that made it more likely that the effects that were anticipated in the 1996 announcement would eventually materialize. Likewise, we expect to find negative treatment estimates if the information made it less likely.

Fig. 8 illustrates the treatment effects estimated for different hypothetical announcement dates. Before the announcement, the

¹² We run a series of q estimations of the following type: $\log(P_{it}) = \theta_{0q}T_i + \theta_{1q}(T_i \times POST96_t) + \beta_q POST96_t + \gamma_q TREND_t + \sum_o b_{oq}X_{io} + \mu_{iq} + e_{qit}$. In each estimation, we use a window of observations of one year before and one year after the intervention date.

Table 7

Support for the new airport concept: Interaction effects of homeownership rate and the 1996 and 2007 price signals.

	(1) NO votes [percent] REF2008	(2) NO votes [percent] REF2008	(3) NO votes [percent] REF2008	(4) NO votes [percent] PLACEBO ^a	(5) NO votes [percent] PLACEBO ^a	(6) NO votes [percent] PLACEBO ^a
Homeownership rate [percent]	0.126*** (0.036)	0.110*** (0.024)	0.111*** (0.022)	0.022 (0.017)	0.037*** (0.015)	0.059*** (0.018)
2007 Price signal [percent]	−0.233 (0.225)	−0.204 (0.210)	−0.708 (2.540)	0.229** (0.111)	0.293*** (0.088)	−1.559 (1.846)
2007 Price signal × Homeownership rate	0.015*** (0.005)	0.012*** (0.003)	0.008** (0.003)	−0.006*** (0.002)	−0.004** (0.002)	−0.001 (0.002)
1996 Price signal [percent]		−0.210 (0.175)	0.190 (2.418)		−0.122 (0.077)	1.205 (1.764)
1996 Price signal × Homeownership rate		0.008** (0.004)	0.009** (0.004)		−0.000 (0.002)	0.001 (0.003)
Controls	YES	YES	YES	YES	YES	YES
Airport variables	NO	NO	YES	NO	NO	YES
N	1201	1201	1201	1201	1201	1201
R ²	0.890	0.893	0.919	0.928	0.929	0.933

Notes: OLS estimates. NO votes is the share of NO votes among voters who voted in person. REF2008 denotes the 2008 referendum on the continued operation of Tempelhof airport. PLACEBO denotes the 2009 referendum on whether ethics should become a compulsory school subject. Standard errors in parentheses are bootstrapped with 500 iterations. All models include a dummy variable to differentiate between former East and West Berlin. Controls and Airport effects are defined in the notes to Table 5. The 2007 price signal has been constructed in perfect analogy to the 1996 price signal with January 1, 2007, as the announcement date.

^a As with REF2008, the voting data for the PLACEBO referendum in 2009 were provided by the Statistical Office for Berlin–Brandenburg.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

1996 price signal seems, if anything, to be negatively correlated with property price trends. Volatility in the estimated parameters is high. After the 1996 announcement, little correlation exists between the 1996 price signal and property price trends for a couple of years. As 2006 approaches, volatility again increases. Using January 1, 2007, as an announcement date, we find a statistically significant adjustment that is positively correlated with the 1996 announcement. This date coincides with the funded decision in favor of the shutdown of Tempelhof airport in August 2006, which was first challenged but, after another revision, eventually passed the Federal Administrative Court of Germany as the court of last instance.

We thus construct a price signal for the January 1, 2007, announcement by using exactly the same procedure as for the 1996 announcement. In Table 7, columns (1–3), we use the resulting price signal instead of and in addition to the 1996 price signal to explain the voting behavior in the Tempelhof referendum. Even conditional on the 1996 effect, the interaction effects turn out to be slightly larger for the 2007 price signal than for the 1996 price signal. This result is consistent with the shorter period that residents had to relocate in response to the information released, which should amplify homevoter and leasevoter effects. The 1996 interaction effects once more pass the placebo tests, as shown in columns (4–6). While the placebo tests of the 2007 composite consumption and wealth effects are less favorable, as evident from columns (4) and (5), the pure 2007 wealth effect passes the placebo test—see column (6).

6. Conclusion

The homevoter hypothesis states that homeowners vote in favor of initiatives perceived to increase and vote against those perceived to decrease the value of their properties (Fischel, 2001). Our analysis adds to the empirical literature supporting this hypothesis (Brunner and Sonstelie, 2003; Brunner et al., 2001; Dehring et al., 2008; Hilber and Mayer, 2009). We enrich the literature by explicitly distinguishing between homevoters, i.e., voters who own their homes, and leasevoters, i.e., voters who lease their homes. We demonstrate that even in a regulatory environment with restrictions on rent increases from sitting renters, as in US cities such as New York or San Francisco, leasevoters and homevoter attitudes toward similar amenity changes point into

the opposite directions. In areas where renters are not similarly protected against rent increases, home- and leasevoter behavior could diverge even more because renters would have greater incentives to oppose rental price effects associated with positive amenity changes.

Our results, and more generally the evidence provided in the related literature, have important implications for the allocation of public facilities with localized effects and particularly for their allocation via direct democracy processes. Our results clearly suggest that individuals have diverse preferences concerning (public) facilities with localized effects, which may imply that decisions regarding such facilities should be based on social cost–benefit analyses, especially those based on revealed preference approaches (Osborne and Turner, 2010). In the long run, a political process based on referenda will not necessarily lead to the allocation of local public goods according to their welfare impact. Instead, it could lead to a concentration of public facilities with negative local externalities (local public bads) in areas that are dominated by renters, whereas homeowners may attract facilities with positive local externalities (local public goods).

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jue.2015.03.002>.

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